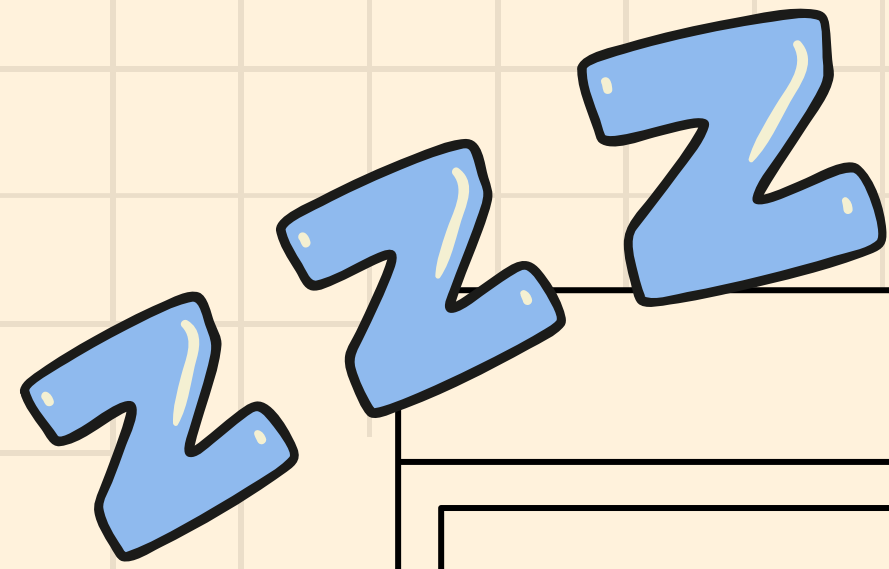


# Sleep Quality & Lifestyle

## A Multiple Linear Regression Analysis

Hannah Athirah



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Survey-based information on health, lifestyle, and sleep characteristics

Sample size: 374 individuals

Variables: 13 (one responding and eight predictors)

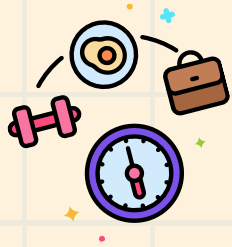
Target Variable: Quality of Sleep, scored from 1 (Poor) to 10 (Excellent)



Where the Data Comes From

Source: Kaggle – Sleep Health and Lifestyle Dataset

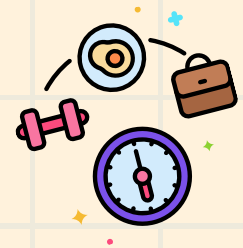
| sleep_data |        |     |                      |                |                  |                         |              |              |                |            |             |                |
|------------|--------|-----|----------------------|----------------|------------------|-------------------------|--------------|--------------|----------------|------------|-------------|----------------|
| Person ID  | Gender | Age | Occupation           | Sleep Duration | Quality of Sleep | Physical Activity Level | Stress Level | BMI Category | Blood Pressure | Heart Rate | Daily Steps | Sleep Disorder |
| 1          | Male   | 27  | Software Engineer    | 6.1            | 6                | 42                      | 6            | Overweight   | 126/83         | 77         | 4200        | None           |
| 2          | Male   | 28  | Doctor               | 6.2            | 6                | 60                      | 8            | Normal       | 125/80         | 75         | 10000       | None           |
| 3          | Male   | 28  | Doctor               | 6.2            | 6                | 60                      | 8            | Normal       | 125/80         | 75         | 10000       | None           |
| 4          | Male   | 28  | Sales Representative | 5.9            | 4                | 30                      | 8            | Obese        | 140/90         | 85         | 3000        | Sleep Apnea    |
| 5          | Male   | 28  | Sales Representative | 5.9            | 4                | 30                      | 8            | Obese        | 140/90         | 85         | 3000        | Sleep Apnea    |
| 6          | Male   | 28  | Software Engineer    | 5.9            | 4                | 30                      | 8            | Obese        | 140/90         | 85         | 3000        | Insomnia       |
| 7          | Male   | 29  | Teacher              | 6.3            | 6                | 40                      | 7            | Obese        | 140/90         | 82         | 3500        | Insomnia       |
| 8          | Male   | 29  | Doctor               | 7.8            | 7                | 75                      | 6            | Normal       | 120/80         | 70         | 8000        | None           |
| 9          | Male   | 29  | Doctor               | 7.8            | 7                | 75                      | 6            | Normal       | 120/80         | 70         | 8000        | None           |
| 10         | Male   | 29  | Doctor               | 7.8            | 7                | 75                      | 6            | Normal       | 120/80         | 70         | 8000        | None           |



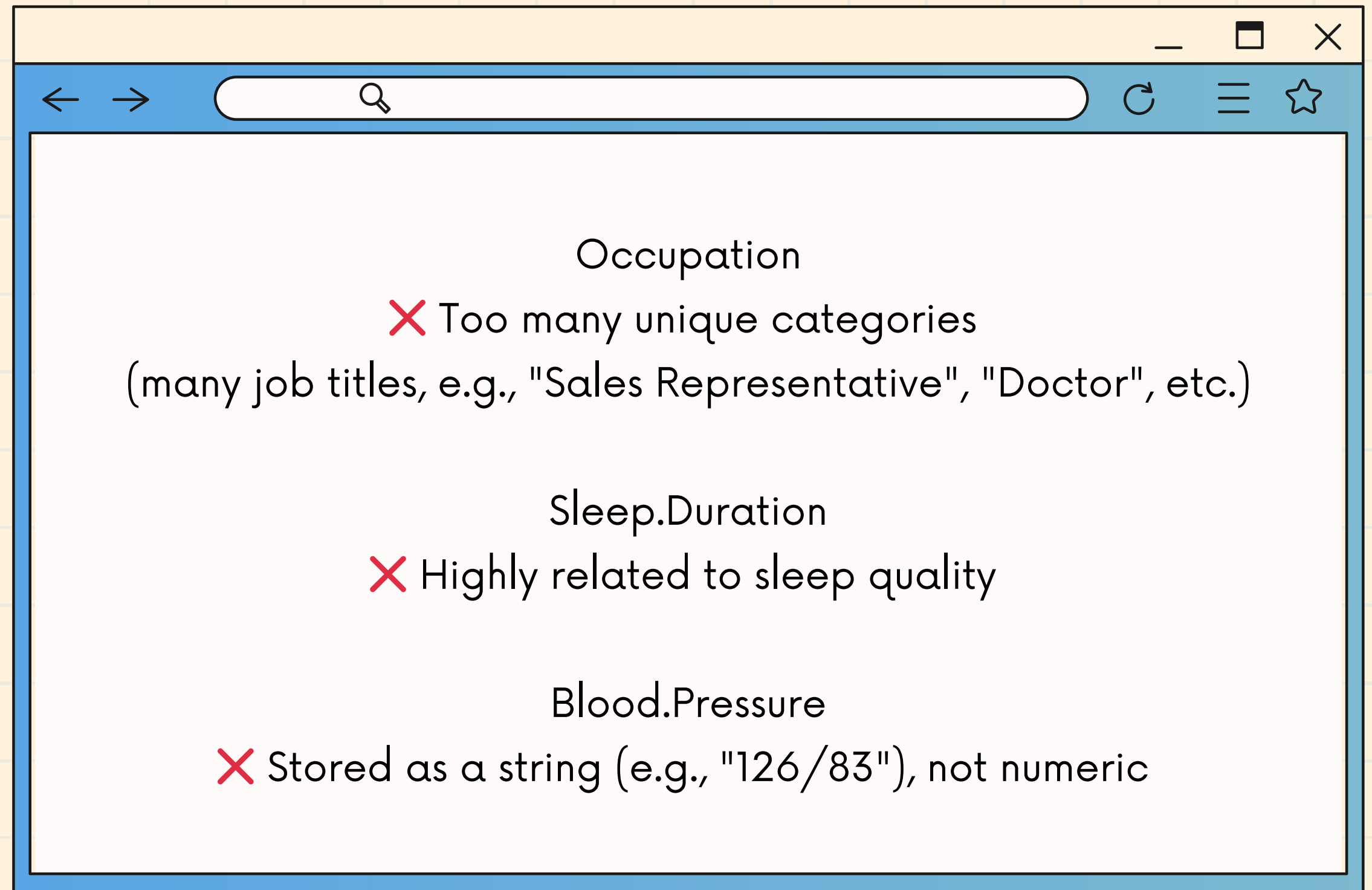
**What lifestyle  
and health  
factors are  
most strongly  
associated  
with sleep  
quality?**

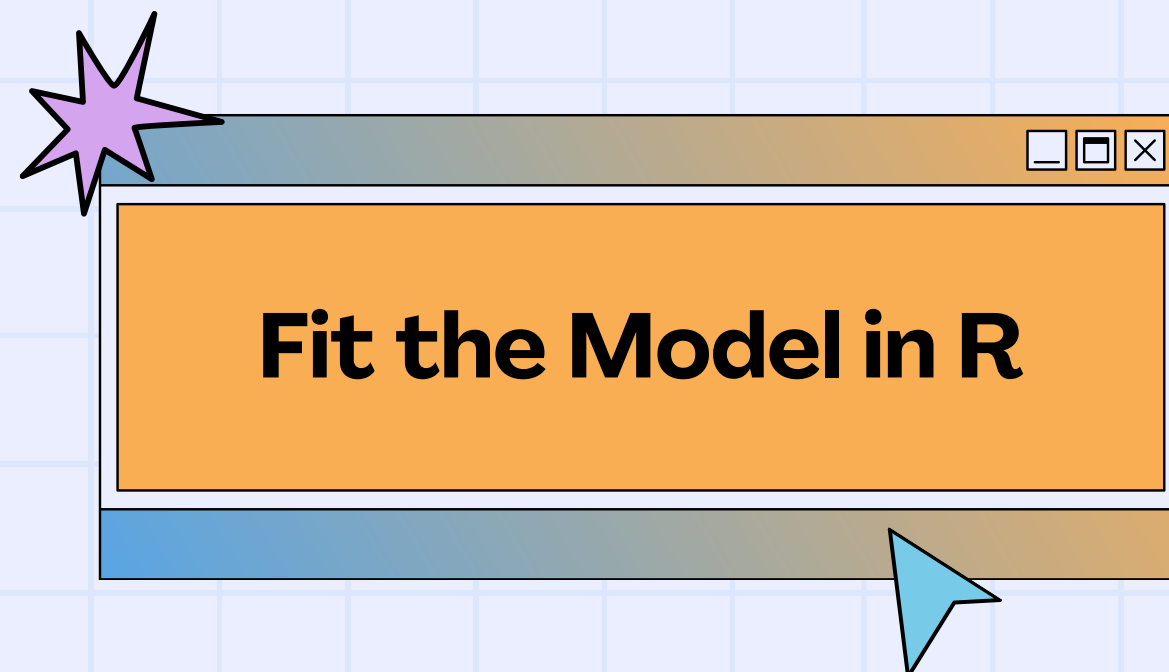


| Type                                                                                                         | Variables                                                                                                            |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Response Variable                                                                                            | Quality of Sleep (scale: 1–10)                                                                                       |
|  Numerical Predictors   | Age, Stress Level, Physical Activity, Heart Rate, Daily Steps                                                        |
|  Categorical Predictors | BMI Category (Underweight, Normal, Overweight, Obese); Sleep Disorder (None, Insomnia, Apnea); Gender (Female, Male) |



**What lifestyle  
and health  
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quality?**





```
1 install.packages("car") # for dWT test
2 library(car)
3
4 data = read.csv("sleep_data.csv")
5 attach(data)
6 head(data)
7
8 # converting categorical variable to factors
9 Gender = factor(Gender)
10 BMI.Category = factor(BMI.Category)
11 Sleep.Disorder = factor(Sleep.Disorder)
12
13 # fitting MLR
14 mlr = lm(Quality.of.Sleep ~ Age + Gender + Stress.Level +
15         Physical.Activity.Level + Heart.Rate + Daily.Steps +
16         BMI.Category + Sleep.Disorder)
17 summary(mlr)
```

```
> summary(mlr)
```

Call:

```
lm(formula = Quality.of.Sleep ~ Age + Gender + Stress.Level +
    Physical.Activity.Level + Heart.Rate + Daily.Steps + BMI.Category +
    Sleep.Disorder)
```

Residuals:

|  | Min      | 1Q       | Median   | 3Q      | Max     |
|--|----------|----------|----------|---------|---------|
|  | -1.03790 | -0.14509 | -0.01048 | 0.12429 | 1.07840 |

Coefficients:

|                           | Estimate   | Std. Error | t value | Pr(> t ) |     |
|---------------------------|------------|------------|---------|----------|-----|
| (Intercept)               | 7.516e+00  | 6.335e-01  | 11.866  | < 2e-16  | *** |
| Age                       | 5.594e-02  | 2.796e-03  | 20.004  | < 2e-16  | *** |
| GenderMale                | 3.147e-01  | 4.330e-02  | 7.267   | 2.27e-12 | *** |
| Stress.Level              | -4.740e-01 | 2.111e-02  | -22.459 | < 2e-16  | *** |
| Physical.Activity.Level   | 3.896e-03  | 1.643e-03  | 2.371   | 0.018267 | *   |
| Heart.Rate                | -7.257e-03 | 9.729e-03  | -0.746  | 0.456227 |     |
| Daily.Steps               | 3.025e-05  | 2.157e-05  | 1.402   | 0.161671 |     |
| BMI.CategoryNormal Weight | -5.802e-02 | 7.004e-02  | -0.828  | 0.407971 |     |
| BMI.CategoryObese         | -6.650e-01 | 1.649e-01  | -4.032  | 6.75e-05 | *** |
| BMI.CategoryOverweight    | -7.422e-01 | 5.830e-02  | -12.729 | < 2e-16  | *** |
| Sleep.DisorderNone        | 2.895e-01  | 5.732e-02  | 5.051   | 6.99e-07 | *** |
| Sleep.DisorderSleep Apnea | 2.296e-01  | 5.912e-02  | 3.884   | 0.000122 | *** |
| ---                       |            |            |         |          |     |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2841 on 362 degrees of freedom

Multiple R-squared: 0.9453, Adjusted R-squared: 0.9437

F-statistic: 568.9 on 11 and 362 DF, p-value: < 2.2e-16

## The Model & Its Interpretation

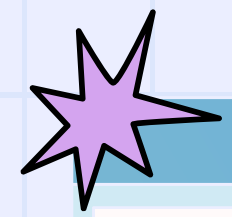
Residual standard error: 0.2841 on 362 degrees of freedom  
Multiple R-squared: 0.9453, Adjusted R-squared: 0.9437  
F-statistic: 568.9 on 11 and 362 DF, p-value: < 2.2e-16

Adjusted R<sup>2</sup>: 0.944  
→ The model explains 94.4% of the variation in sleep quality — very strong fit

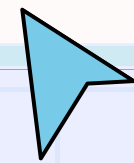
Residual Standard Error: 0.284  
→ Predictions are typically off by ~0.28 units — highly precise

F-statistic: 568.9, p < 2.2e-16  
→ The model is **statistically significant overall**

$$\begin{aligned}\hat{Y}_{\text{Sleep Quality}} = & 7.516 \\ & + 0.056 \cdot \text{Age} \\ & + 0.315 \cdot \text{Male} \\ & - 0.474 \cdot \text{Stress Level} \\ & + 0.0039 \cdot \text{Physical Activity} \\ & - 0.0073 \cdot \text{Heart Rate} \\ & + 0.00003 \cdot \text{Daily Steps} \\ & - 0.665 \cdot \text{Obese} \\ & - 0.742 \cdot \text{Overweight} \\ & + 0.290 \cdot \text{No Sleep Disorder} \\ & + 0.230 \cdot \text{Sleep Apnea}\end{aligned}$$



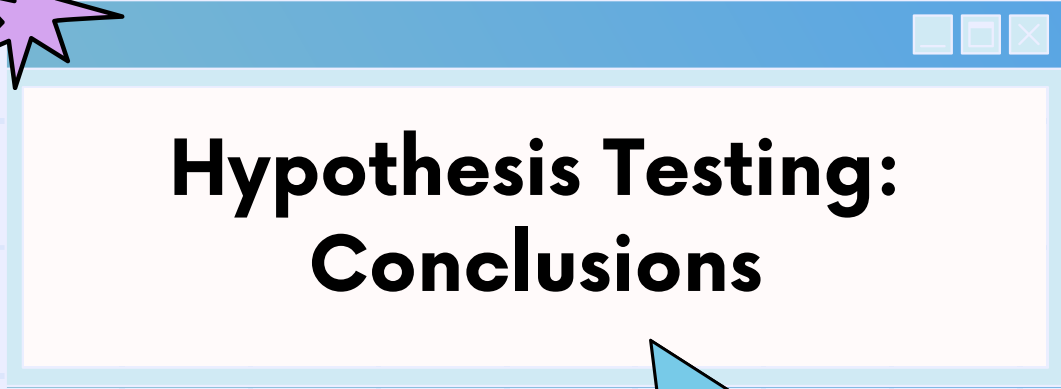
## Hypothesis Testing: Hypotheses & Decisions



$H_0 : \beta_i = 0$  no effect on  
sleep quality

$H_a : \beta_i \neq 0$  has an effect on  
sleep quality

| Significance                                                                                                        |                                                                                       | Significance Level Used: $\alpha = 0.05$                                                                               |                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <br>$p < 0.05$ ,<br>Reject $H_0$ |  | <ul style="list-style-type: none"><li>• Age</li><li>• BMI: Obese / Overweight</li><li>• Sleep Disorder: None</li></ul> | <ul style="list-style-type: none"><li>• Sleep Apnea</li><li>• Gender: Male</li><li>• Physical Activity</li><li>• Stress Level</li></ul> |
|                                                                                                                     |                                                                                       | <ul style="list-style-type: none"><li>• Heart Rate</li><li>• Steps</li></ul>                                           |                                                                                                                                         |



## Hypothesis Testing: Conclusions

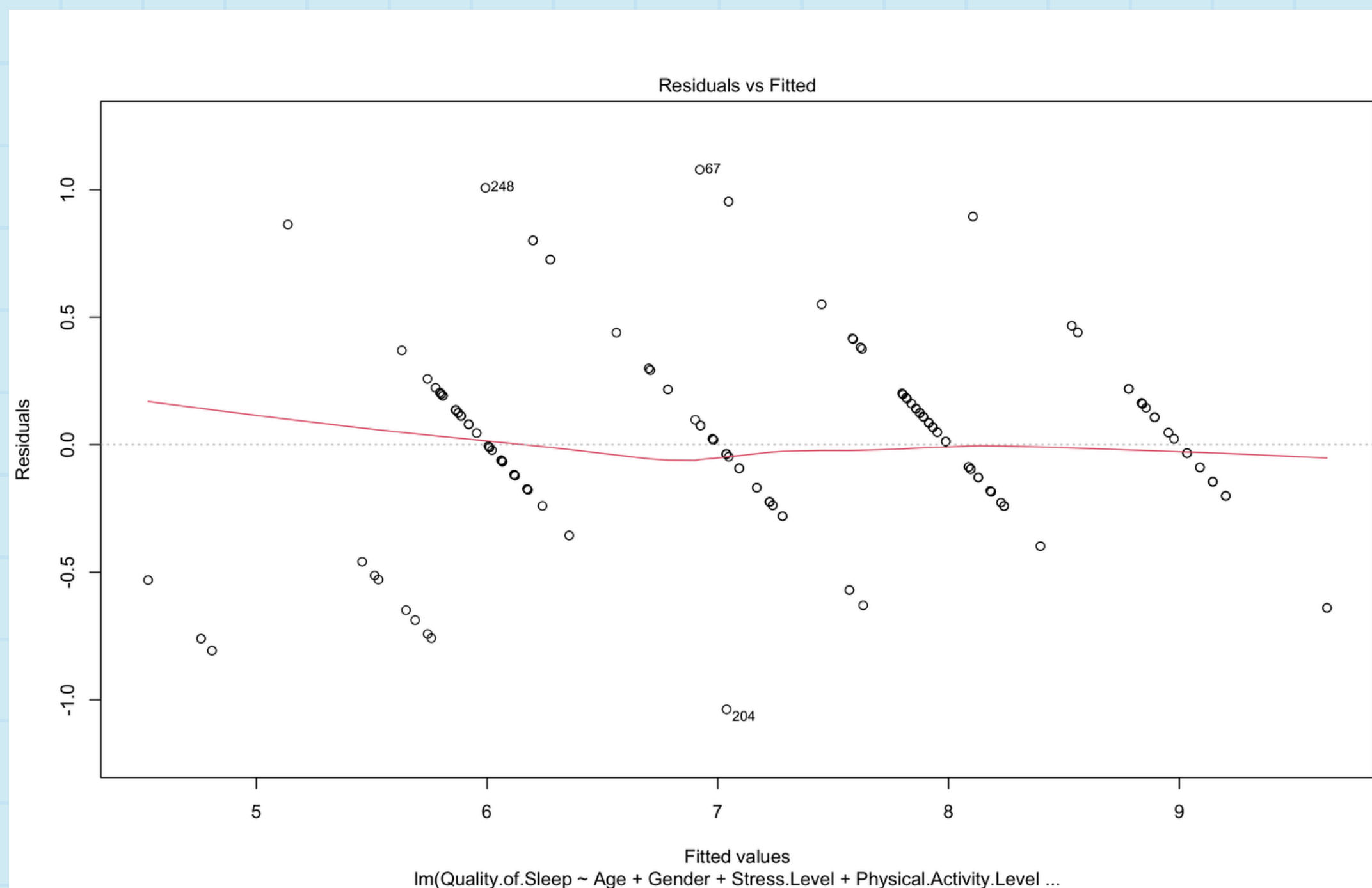


- **Stress level** had the largest negative effect on sleep quality.
- Individuals who are **male, not obese**, and **have no sleep disorder** report better sleep quality than the baseline group.
- The baseline group includes:
  - **Female individuals**
  - **People with insomnia**
  - **Normal BMI category**
- **Age** was associated with slightly better sleep, while **Heart Rate** and **Daily Steps** were not statistically significant predictors.

## Assumption Check #1: Linearity

- ✓ Residuals are scattered randomly around 0
- ✓ No strong curve or U-shape
- ✓ The red smoothing line is fairly flat
- ✓ No visible trend or systematic pattern

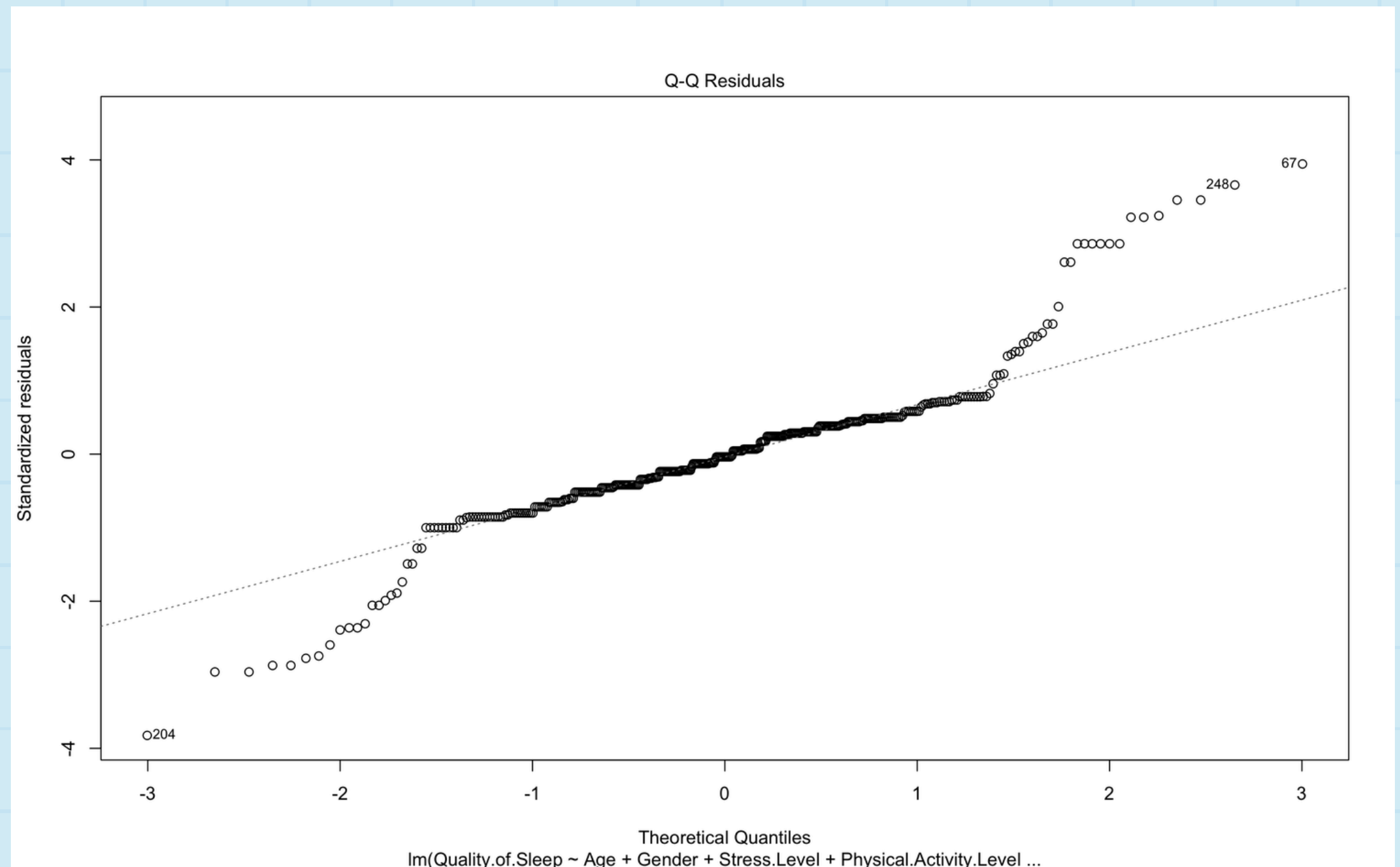
```
> plot(mlr, which = 1) # Linearity: Residuals vs Fitted
```



## Assumption Check #2: Normality

- ✓ Most of the points fall along the diagonal reference line
- ✓ Slight deviations at the tails are acceptable in larger samples
- ✓ No severe departures from normality

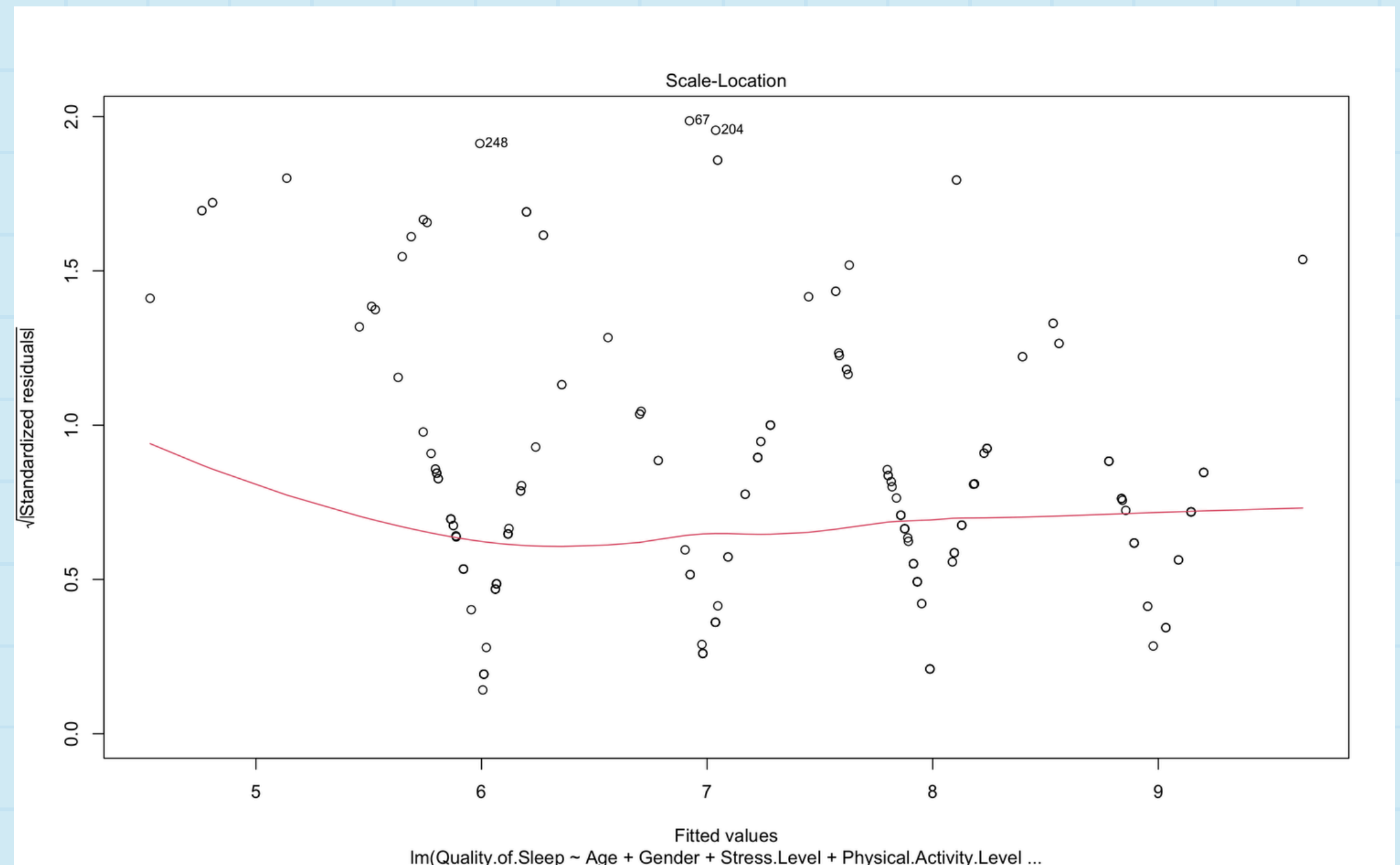
```
> plot(mlr, which = 2) # Normality: Q-Q plot
```



### Assumption Check #3: Homoscedasticity

- ✓ No clear funnel shape
- ✓ The red line is mostly flat, indicating constant variance
- ✓ Points are scattered fairly evenly — no visible pattern
- ✓ Suggests the equal variance assumption is satisfied

```
> plot(mlr, which = 3) # Homoscedasticity: Equal Variance
```



## Assumption Check #4: Autocorrelation

```
> durbinWatsonTest(mlr) # Independence Assumed
lag Autocorrelation D-W Statistic p-value
1      0.3862648      1.225369      0
Alternative hypothesis: rho != 0
```

⚠ DW = 1.225 < ~2.000

→ Some autocorrelation detected

✓ p-value < 0.05

→ positive autocorrelation

## Multicollinearity Check

```
> vif(mlr) # Multicollinearity
              GVIF Df GVIF^(1/(2*Df))
Age           2.717988 1      1.648632
Gender        2.171324 1      1.473541
Stress.Level   6.481453 1      2.545870
Physical.Activity.Level 5.414210 1      2.326846
Heart.Rate     7.480515 1      2.735053
Daily.Steps    5.627069 1      2.372144
BMI.Category   10.719367 3      1.484892
Sleep.Disorder  5.094803 2      1.502387
```

VIF < 5 → No concern

5 < VIF < 10 → Moderate multicollinearity

VIF > 10 → Serious concern



**Thank you! :)**

#### **References:**

<https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset>